## CONTACT


EMAIL
anupmulay1@gmail.com

WEBSITE
anupmulay.github.io

LINKEDIN
linkedin.com/in/anupmulay1

NATIONALITY
Indian


## LANGUAGE SKILLS

MOTHER TONGUES
Hindi, Marathi

OTHER LANGUAGES

| | Listen | Read | Speak | Interact | Write |
|---|---|---|---|---|---|
| English | C2 | C2 | C2 | C2 | C2 |
| Lithuanian | B1 | B1 | B1 | B1 | B1 |

A1 and A2: Basic user. B1 and B2: Independent user. C1 and C2: Proficient user.

## DIGITAL SKILLS

MATLAB
R
Stata
Python
LaTeX
Microsoft Excel

## PROFESSIONAL SKILLS

Structural and semistructural modelling
DSGE models
Heterogeneous agent models
Macroeconometrics
Microeconometrics (panel data)
Firm level microdata analysis
Computational methods
Forecasting and policy scenarios

# Anup Mulay


Principal Economist, Lietuvos bankas

Applied Macroeconomic Research Division | Vilnius, Lithuania


## ABOUT ME

I am a Principal Economist in the Applied Macroeconomic Research Division at Lietuvos bankas, where I specialise in structural and semistructural macroeconomic modelling and its application to policy analysis.

## WORK EXPERIENCE

### Principal Economist
02/2025 – Current

Lietuvos bankas | Vilnius, Lithuania

- Responsible for developing several of the Bank's core macroeconomic models.
- Lead the Bank's policy scenario work, from designing scenarios to delivering results, on a wide range of macroeconomic questions, from external shocks such as commodity price swings to fiscal policy and structural reforms.
- Hold ownership of, and responsibility for, policy analysis and model development tasks in the division.
- Independently built E-LITMAS, the version of LITMAS the Bank uses for energy and climate policy, with a full energy sector that models fossil fuels alongside green and brown energy production, and used it for energy price shock and climate transition scenarios for the Baltic economies.
- Continue to develop, estimate and update LITMAS jointly with Stefan Girstmair.
- Maintain, update and modify NMCM, the Bank's primary semistructural forecasting tool. NMCM and LITMAS feed into the Bank's regular forecasting rounds.

### Senior Economist
10/2021 – 02/2025

Lietuvos bankas | Vilnius, Lithuania

- Built LITMAS (LIThuanian Model for Analysis of Scenarios), a large scale estimated DSGE model, jointly with Stefan Girstmair, and estimated, modified and updated it.
- Started development of E-LITMAS, the energy sector extension of LITMAS.
- Maintained, updated and modified NMCM, the Bank's primary semistructural forecasting tool, as input to the Bank's regular forecasting rounds.
- Contributed to the Bank's policy scenario work as a member of the scenario team.

### Lecturer
Spring 2023, 2025, 2026

Vilnius University, Faculty of Economics and Business | Vilnius, Lithuania

- Taught Economic Theory II, a second year bachelor's course in intermediate macroeconomic theory covering the Solow growth model, the overlapping generations model, dynamic consumption and saving, IS LM, a full production economy with fiscal policy and Ricardian equivalence, and the role of money.

### Postdoctoral Researcher
03/2021 – 10/2021

University of Kent | Canterbury, United Kingdom

Investigated trends in market power and their relationship with aggregate productivity using UK Data Service Business Microdata, with the objective of building structural models to explain these phenomena and generate policy implications.

### Research Assistant
04/2019 – 08/2019

University of Kent | Canterbury, United Kingdom

- Research assistant to Dr Anthony Savagar.
- Replicated the DSGE model of Jaimovich and Floetotto (2008) in MATLAB.
- Carried out extensive microeconometric and panel data analysis in Stata and R.
- Decomposed TFP from data using various specifications in R.
- Compared results across specifications numerically and compiled the findings in LaTeX.

### Teaching Assistant
09/2015 – 12/2019

University of Kent | Canterbury, United Kingdom

- **EC562 Financial Economics:** martingale option pricing, the HJB method for option pricing, the binomial model (algebra and MATLAB), interest rate models, Monte Carlo simulations and binomial asset pricing in MATLAB and Excel.

- **EC534 Money and Banking:** theoretical models of money and monetary policy (core text: Champ, Freeman and Haslag, 4th edition).
- **EC314 Data Analysis:** linear programming in Excel, downloading data, HP filtering, regression analysis and report generation.

**Research Assistant** 07/2017 – 11/2017
National Institute of Economic and Social Research (NIESR) | London, United Kingdom

- Member of the team building a nowcasting model that produces quarterly estimates of economic variables.
- Data cleaning and analysis in Python.
- Transferred and rewrote the model code from an older version of MATLAB to the current version, with modifications.

**Researcher** 01/2015 – 03/2015
Bridge2Tech Pvt Ltd | Pune, India

- Conducted secondary research on developing economies.
- Analysed the investment potential of countries in terms of fiscal and monetary policy and their medium and long term outlook.
- Presented the results as slide decks.

**Article Trainee (Chartered Accountancy)** 05/2004 – 07/2008
S.S. Mulay and Co., Chartered Accountants | Pune, India
Mandatory article clerk training for the Chartered Accountant qualification: auditing, accounting, report generation and tax return filing.

## EDUCATION AND TRAINING

**PhD in Economics** 2015 – 2020
University of Kent | Canterbury, United Kingdom
Research on financial frictions and their effect on measured aggregate productivity.

**Graduate studies in Economics** 2013 – 2014
Boston College | Boston, United States
Also worked as a research assistant during the year.

**Master's degree in Economics** 2011 – 2012
University of Nottingham | Nottingham, United Kingdom

**Master's degree in Investments and Securities** 2009 – 2010
London South Bank University | London, United Kingdom

**Chartered Accountant, Accounting and Finance** 2004 – 2008
The Institute of Chartered Accountants of India | India

## PUBLICATIONS

**An Ex-Ante Assessment of the Proposal to Reform the Second Pillar of the Lithuanian Pension System**
With Karolis Bielskis, Federica Brenna, Jose Garcia-Louzao and Stefan Girstmair. Lietuvos bankas Occasional Paper No. 57/2025. Link

## WORK IN PROGRESS

**Capital Portfolio, Collateral Constraints and the Measured Efficiency Wedge**
With Alfred Duncan. SSRN

**An E-DSGE for the Baltics**
With Patrick Grüning.

**The Welfare Costs of Distortions with Imperfectly Contestable Markets**
With Anthony Savagar.

## PROJECTS

**The Demography of Currency Unions**
With Swapnil Singh. In progress. A two country HANK OLG model of a currency union whose members age at different rates. The model combines an exact demographic transition with a structural general equilibrium block to study how this cross country demographic heterogeneity shapes the transmission of a common monetary policy.